SmartNest AI

Problem Statement:

Finding rental houses is still a difficult, time-consuming, and unreliable process in many cities and districts. Most people depend heavily on brokers or third-party agents to search for rental properties, which often leads to high brokerage fees, fake listings, misleading property information, lack of transparency, and wasted time visiting unavailable or unsuitable houses.

Existing rental platforms mainly focus on basic property listings and do not provide intelligent verification, real-time availability tracking, locality analysis, or personalized recommendations. Users also face challenges in understanding the actual condition of a property because uploaded images and descriptions may be outdated, manipulated, or incomplete. In many cases, rental houses are booked before interested users can physically visit them, creating inconvenience and competition among renters.

Additionally, there is no efficient system that combines:

- real-time map-based property visualization,
- AI-based property quality analysis,
- fake listing detection,
- virtual property inspection,
- intelligent rental recommendations,
- temporary booking/hold mechanisms.

Therefore, there is a need for an intelligent rental ecosystem analysis that can help users discover verified rental houses directly through an interactive map interface while reducing dependency on brokers and improving trust, transparency, and convenience in the rental process.

The proposed system aims to develop a Multimodal AI-Based Smart Rental Platform that allows property owners or verified agents to upload rental house details, images, and videos, while AI models analyze the uploaded multimedia content to evaluate property conditions, detect fake listings, estimate fair rental prices, and provide smart recommendations to users based on their preferences and location.

The system will also support:

- map-based rental exploration,
- voice-based property search,
- temporary house holding/booking,
- real-time availability updates,
- and locality intelligence analysis.

By integrating Artificial Intelligence, Machine Learning, Computer Vision, NLP, Audio Processing, and Geospatial Mapping, the proposed platform seeks to create a secure, transparent, and intelligent rental house discovery solution for modern users.